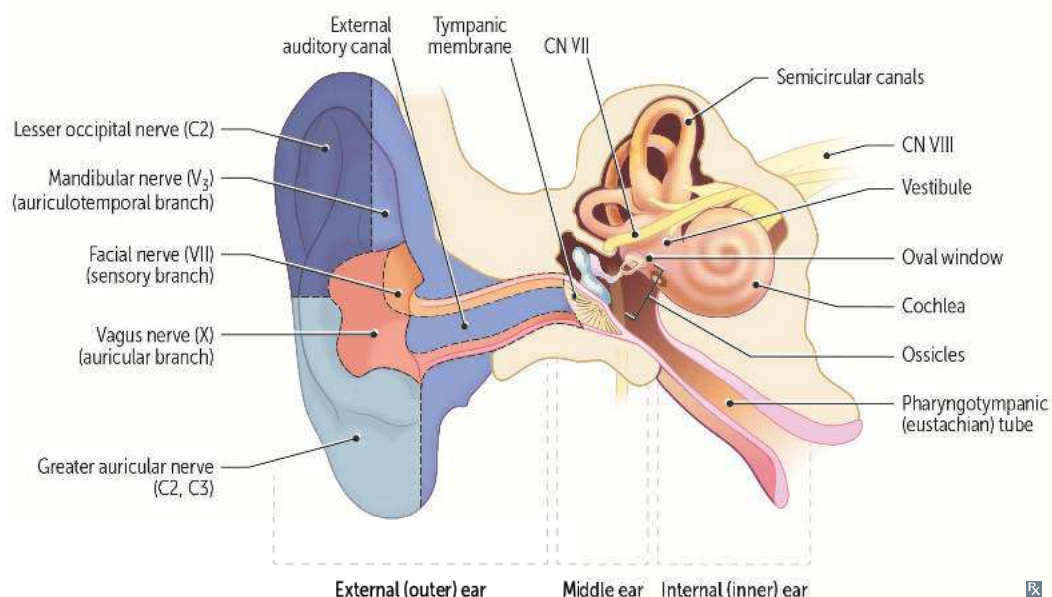


► NEUROLOGY—OTOLOGY

Auditory anatomy and physiology

Outer ear	Visible portion of ear (pinna), includes auditory canal and tympanic membrane. Transfers sound waves via vibration of tympanic membrane.
Middle ear	Air-filled space with three bones called the ossicles (malleus, incus, stapes). Ossicles conduct and amplify sound from tympanic membrane to inner ear.
Inner ear	Snail-shaped, fluid-filled cochlea. Contains basilar membrane that vibrates 2° to sound waves. Vibration transduced via specialized hair cells → auditory nerve signaling → brainstem. Each frequency leads to vibration at specific location on basilar membrane (tonotopy): ▪ Low frequency heard at apex near helicotrema (wide and flexible). ▪ High frequency heard best at base of cochlea (thin and rigid).

Otitis externa

Inflammation of external auditory canal. Most commonly due to *Pseudomonas*. Associated with water exposure (swimmer's ear), ear canal trauma/occlusion (eg, hearing aids). Presents with otalgia that worsens with ear manipulation, pruritus, hearing loss, discharge **A**. **Malignant (necrotizing) otitis externa**—invasive infection causing osteomyelitis. Complication of otitis externa mostly seen in older patients with diabetes. Presents with severe otalgia and otorrhea. May lead to cranial nerve palsies. Physical exam shows granulation tissue in ear canal.

Otitis media

Inflammation of middle ear. Most commonly due to nontypeable *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis*. Associated with eustachian tube dysfunction, which promotes overgrowth of bacterial colonizers of upper respiratory tract. Usually seen in children < 2 years old. Presents with fever, otalgia, hearing loss. Physical exam shows bulging, erythematous tympanic membrane **A** that may rupture. **Mastoiditis**—infection of mastoid process of temporal bone. Complication of acute otitis media due to continuity of middle ear cavity with mastoid air cells. Presents with postauricular pain, erythema, swelling. May lead to brain abscess.

Common causes of hearing loss

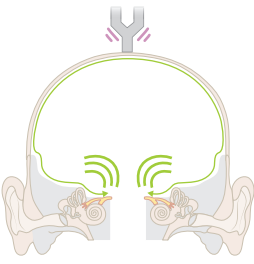
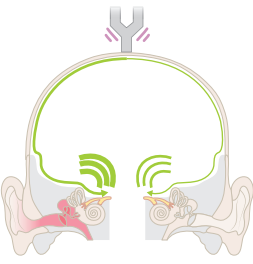
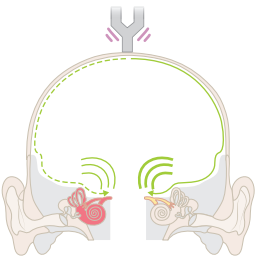
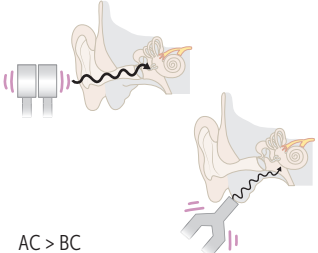
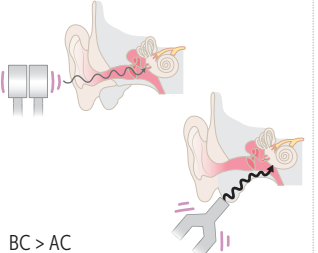
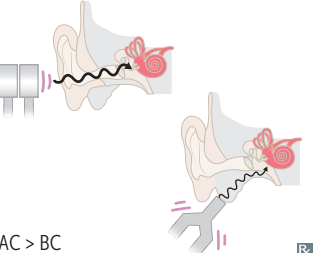
Noise-induced hearing loss

Damage to stereociliated cells in organ of Corti. Loss of high-frequency hearing first. Sudden extremely loud noises can produce hearing loss due to tympanic membrane rupture.

Presbycusis

Aging-related progressive bilateral/symmetric sensorineural hearing loss (often of higher frequencies) due to destruction of hair cells at the cochlear base (preserved low-frequency hearing at apex).

Diagnosing hearing loss

	Normal	Conductive	Sensorineural
Weber test Tuning fork on vertex of skull	 No localization	 Localizes to affected ear ↓ transmission of background noise	 Localizes to unaffected ear ↓ transmission of all sound
Rinne test Tuning fork in front of ear (air conduction, AC), Tuning fork on mastoid process (bone conduction, BC)	 AC > BC	 BC > AC	 AC > BC

Cholesteatoma

Abnormal growth of keratinized squamous epithelium in middle ear **A** (“skin in wrong place”). Usually acquired, but can be congenital. 1° acquired results from tympanic membrane retraction pockets that form due to eustachian tube dysfunction. 2° acquired results from tympanic membrane perforation (eg, due to otitis media) that permits migration of squamous epithelium to middle ear. Classically presents with painless otorrhea. May erode ossicles → conductive hearing loss.

Vertigo

Sensation of spinning while actually stationary. Subtype of “dizziness,” but distinct from “lightheadedness.” Peripheral vertigo is more common than central vertigo.

Peripheral vertigo

Due to inner ear pathologies such as vestibular neuritis; **benign paroxysmal positional vertigo**—semicircular canal debris → episodic vertigo lasting ≤ 1 minute provoked by certain head movements (diagnosed via Dix-Hallpike maneuver, treated via Epley maneuver); and **Ménière disease**—endolymphatic hydrops (↑ endolymph in inner ear) → triad of **vertigo**, **sensorineural** hearing loss, **tinnitus** (“**men** wear **vests**”). Findings: mixed horizontal-torsional nystagmus (never purely torsional or vertical) that does not change direction and is suppressible with visual fixation.

Central vertigo

Due to brainstem or cerebellar lesions (eg, stroke affecting vestibular nuclei, demyelinating disease, or posterior fossa tumor). Findings: nystagmus of any direction that is not suppressible with visual fixation, neurologic findings (eg, diplopia, ataxia, dysmetria).